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International Mathematics Assessments for Schools

2022 /2023 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

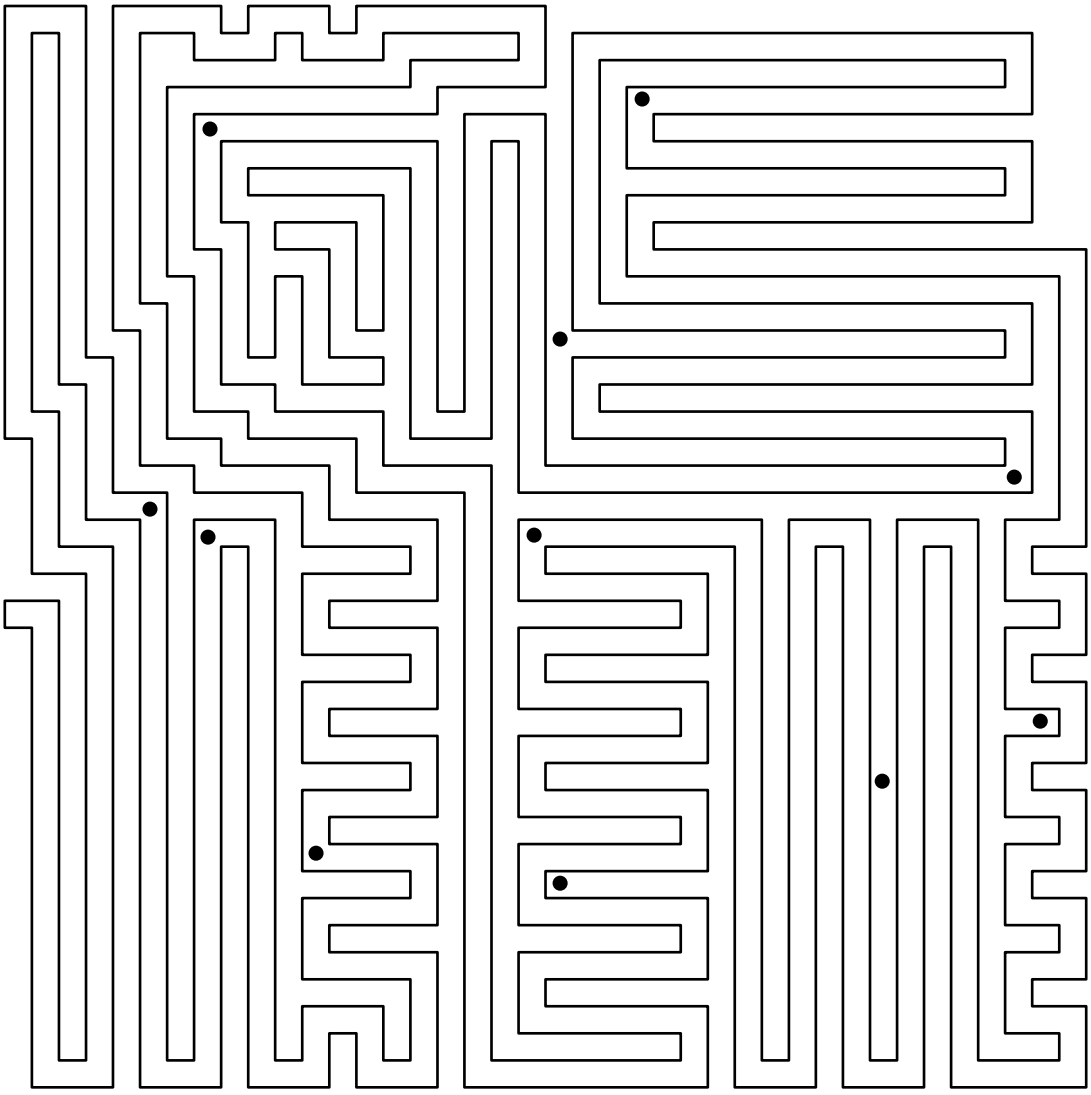
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2022 / 2023 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. The black line forms one closed loop, as shown in the diagram below.



How many dots are there which are located inside the loop?

（A）6　　（B）7　　（C）8　　（D）10　　（E）11

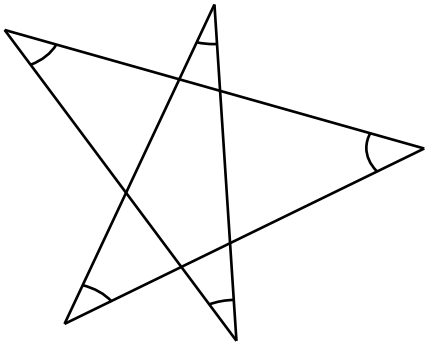
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2. What is the sum of the first two prime numbers that are both greater than 80?

（A）164　　（B）168　　（C）170　　（D）172　　（E）176

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3. As shown in the diagram below, what is the sum, in degrees, of all marked angles?



（A）130°　（B）150°　（C）180°　（D）210°　（E）250°

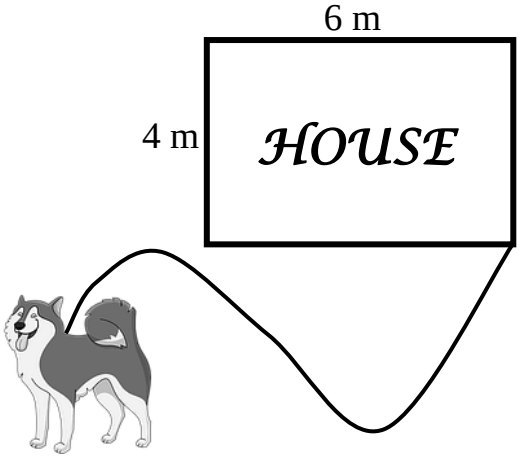
4. After spending $\frac{1}{3}$ of his money and losing $\frac{1}{3}$ of its remainder, Alex now has 120 dollars. How much money, in dollars, did Alex originally have?

（A）180　（B）270　（C）360　（D）540　（E）1080

5. Alice travels from home to school with an average speed of 40 km/h. When she returns home on the same road, she travels with an average speed of 60 km/h. What is her average speed, in km/h, for the entire trip?

（A）45　（B）48　（C）50　（D）52　（E）54

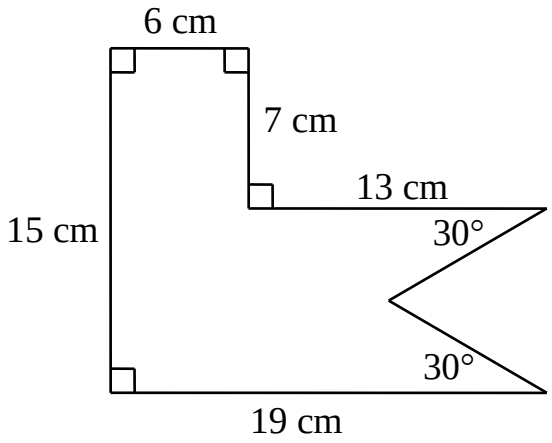
6. A dog is tied to a 10 m inextensible rope in one corner of a rectangular house, as shown in the diagram below. If the length and the width of the house are 6 m and 4 m, respectively, what is the area, in m^2, of the region where the dog can move? (Take $\pi = 3.14$)



（A）235.50　（B）248.06　（C）263.76　（D）276.32　（E）282.60

7. How many 4-digit odd numbers can be formed by using the digits 0, 1, 2, 3 and 4, where repetition of digits is allowed?
（A）120　（B）200　（C）250　（D）500　（E）625

8. A heptagon has five of its sides equal to 6 cm, 7 cm, 13 cm, 15 cm and 19 cm, respectively. It also has three interior right angles, one interior angle of $270°$ and two interior acute angles, each of $30°$, as shown in the diagram below.



What is the perimeter, in cm, of this heptagon?
（A）76　（B）77　（C）78　（D）79　（E）80

9. Find the minimum number of terms in the expression $4+8+12+...$, where the difference between any two consecutive terms is a fixed value, such that their sum exceeds 2022.
（A）30　（B）31　（C）32　（D）33　（E）34

10. In each turn, Alice and Bob alternately picks and crosses out one number from the set: 5, 6, 8, 14, 21, 25, 35, 37 and 63, where Alice goes first. After four turns, the sum of numbers removed by Bob is four times the sum of numbers removed by Alice. What is the remaining number?
（A）8　（B）14　（C）21　（D）25　（E）37

Questions 11-20, 4 marks each

11. There are 8 red, 6 white, 4 green and 2 black balls in a box. What is the least number of balls one needs to draw at random from the box in order to be sure that balls of at least three different colours are drawn?
（A）7　（B）11　（C）14　（D）15　（E）18

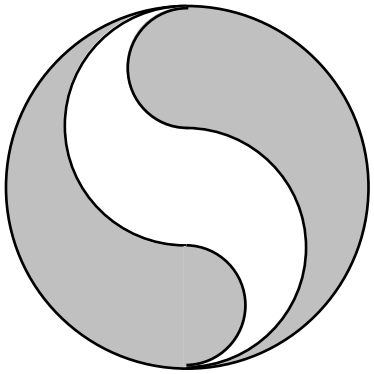
12. The price of an object was increased by p% and then decreased by q% so that the object has the final price equal to the initial one. What is the value of $\frac{pq}{p-q}$?

(A) 80 (B) 120 (C) 1 (D) $\frac{84}{5}$ (E) 100

13. A mutant octopus can have six, seven, eight or nine legs. Those with seven or nine legs always lie, while those with six or eight legs always tell the truth.
The green octopus says: "I have eight legs. The blue octopus has six legs."
The blue octopus says: "I have eight legs. The green octopus has seven legs."
The red octopus says: "I have nine legs. The blue octopus has eight legs."
The yellow octopus says: "I have eight legs. None of the others has eight legs."
Which mutant octopus actually has eight legs?

(A) green octopus (B) blue octopus (C) red octopus
(D) yellow octopus (E) none of them

14. The outer circle has radius 3 cm. The white region is outlined by half circles whose radii are 1 cm and 2 cm and whose centres lie on the same diameter of the outer circle, as shown in the diagram below.



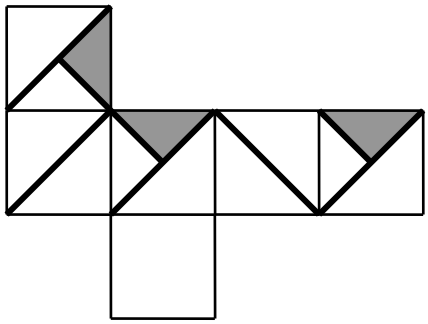
What is the ratio between the areas of shaded region to the white region enclosed by the outer circle?

(A) 2:1 (B) 3:1 (C) 3:2 (D) 8:3 (E) 9:4

15. Using the positive integers from 1, 2, 3, …, 2022, we write each integer one after the other to form one very large number 123456789101112…2022. What is the remainder of this number when divided by 9?

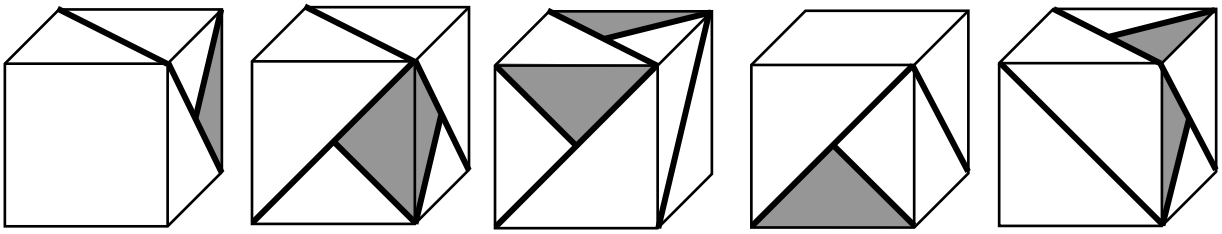
(A) 0 (B) 1 (C) 3 (D) 4 (E) 6

16. The diagram below shows the net of an unfolded cube.



Which of the five cubes below it is not a made-up version of the flattened one?

(A) (B) (C) (D) (E)



17. There are several distinct positive integers such that the sum of any three of them is always a prime number. At most how many such integers?

（A）3 （B）4 （C）5 （D）6 （E）7

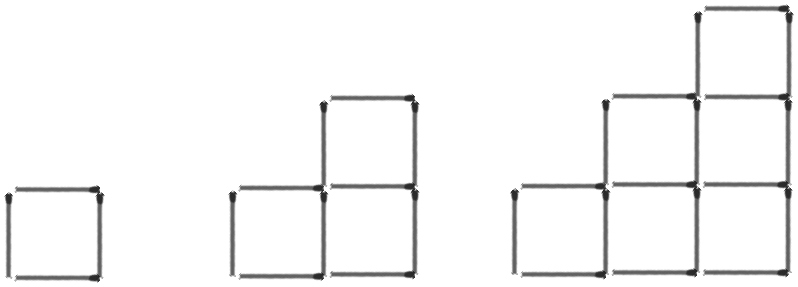
18. Charles has a farm on a 3×3 plot. The farmhouse is in the central lot, and pigs are kept in the eight other lots, with at least one pig in each lot. The total number of pigs in the three lots on any of the four sides of the farm is 9. An example is shown in the diagram below, where the total number of pigs is 28.

3	5	1
5		5
1	5	3

Which of the five numbers below **CANNOT** be the total number of pigs?

（A）20 （B）24 （C）27 （D）32 （E）36

19. The staircase patterns shown below require 4, 10 and 18 matchsticks to construct them.



4 10 18

How many matchsticks are needed to build a staircase of height 100?

（A）10000 （B）10288 （C）10300 （D）10308 （E）11308

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20. In a game, two players take turns calling out any positive integer from 1 to 12. The total of all numbers called out so far is computed. The player who brings the total to 100 or over is the winner. If you go first, you can have a sure win strategy by calling out which number?

（A）1 （B）3 （C）7 （D）9 （E）12

Questions 21-25, 6 marks each

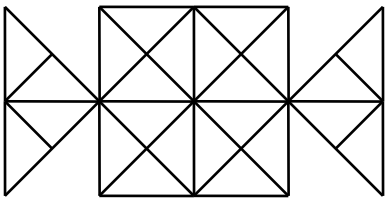
21. In a poll of 500 coffee drinkers, 40% preferred their coffee with neither cream nor sugar and 60% of the remainder preferred their coffee with cream only. After deducting both of these groups, 40% of those left preferred their coffee with sugar only. How many coffee drinkers who preferred coffee with cream and sugar are there?

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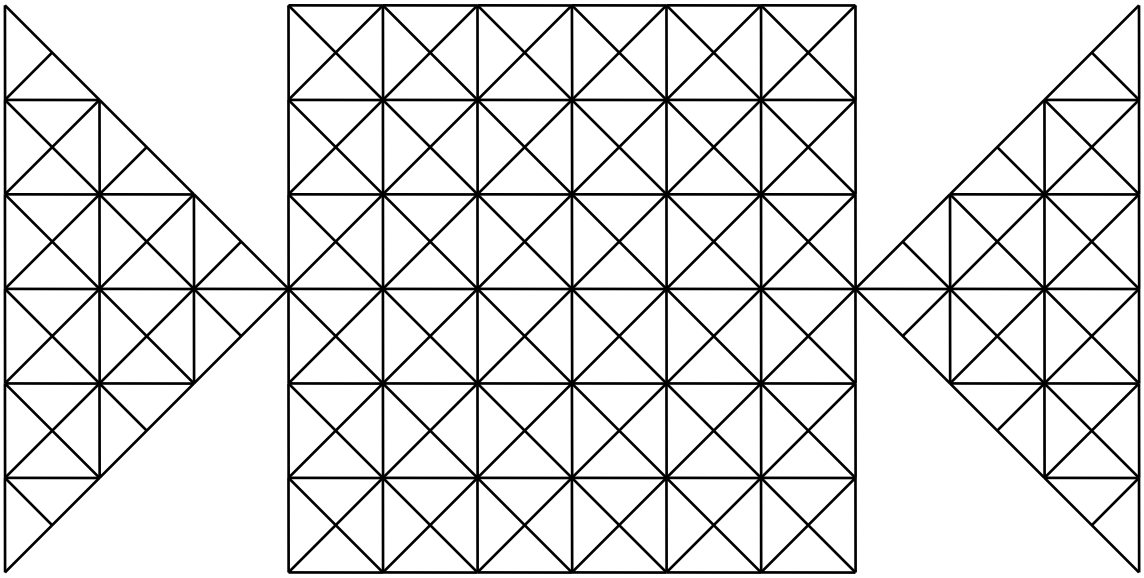
22. Let N be the smallest positive integer whose sum of its digits is 2022. What is the sum of all digits of $N+1$?

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23. The diagram below shows a candy shaped figure.



How many candy shapes with the same size as above can you find within the large grid below?



24. In the grid below, each of its rows and columns contains five numbers which all add up to 100. Fill in the missing numbers to discover the value of the symbol *.

22	31	22	1	24
22				2
23		*		17
17				39
16	4	18	44	18

These are the numbers which have to be filled but they are all jumbled up:

25, 25, 15, 15, 14, 16, 20, 24, 26.

25. How many distinct numbers can be obtained as a product of two or more numbers from the numbers 2, 3, 3, 5, 5, 5, 7, 7, 7 and 7?

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